

|                                          |                                   |                                   |                                   |                                   |                                   |                                   |                                   |                                   |                                   |                                   |                                   |                                   |
|------------------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|
| 2m Temperature (land-only) [celsius]     | 13.42<br><small>13.70</small>     | 2.48<br><small>2.65</small>       | 23.68<br><small>24.24</small>     | 14.72<br><small>14.66</small>     | 5.19<br><small>6.53</small>       | -12.98<br><small>-11.46</small>   | 21.29<br><small>22.70</small>     | 21.31<br><small>20.51</small>     | 20.88<br><small>20.19</small>     | 17.72<br><small>16.59</small>     | 24.75<br><small>24.60</small>     | 7.72<br><small>8.66</small>       |
| Mean Sea Level Pressure [hPa]            | 1011.14<br><small>1011.43</small> | 1016.23<br><small>1014.76</small> | 1012.01<br><small>1013.20</small> | 1004.23<br><small>1004.66</small> | 1011.46<br><small>1011.58</small> | 1017.80<br><small>1015.89</small> | 1011.45<br><small>1012.99</small> | 1005.11<br><small>1004.55</small> | 1010.83<br><small>1011.30</small> | 1013.63<br><small>1012.86</small> | 1013.01<br><small>1013.66</small> | 1003.45<br><small>1005.09</small> |
| Precipitation [mm/day]                   | 2.70<br><small>2.85</small>       | 1.93<br><small>2.15</small>       | 3.37<br><small>3.40</small>       | 2.08<br><small>2.45</small>       | 2.69<br><small>2.80</small>       | 2.04<br><small>2.27</small>       | 3.47<br><small>3.44</small>       | 1.69<br><small>2.06</small>       | 2.78<br><small>2.93</small>       | 1.91<br><small>2.12</small>       | 3.35<br><small>3.40</small>       | 2.46<br><small>2.80</small>       |
| Evaporation [mm/day]                     | -2.71<br><small>-2.91</small>     | -1.47<br><small>-1.74</small>     | -3.78<br><small>-3.95</small>     | -1.71<br><small>-1.99</small>     | -2.70<br><small>-2.87</small>     | -1.51<br><small>-1.76</small>     | -3.87<br><small>-3.99</small>     | -1.43<br><small>-1.74</small>     | -2.80<br><small>-3.01</small>     | -1.47<br><small>-1.76</small>     | -3.83<br><small>-4.03</small>     | -1.95<br><small>-2.20</small>     |
| Precip. minus Evap. [mm/day]             | -0.01<br><small>0.02</small>      | 0.45<br><small>0.53</small>       | -0.41<br><small>-0.47</small>     | 0.37<br><small>0.50</small>       | -0.01<br><small>0.04</small>      | 0.53<br><small>0.61</small>       | -0.40<br><small>-0.45</small>     | 0.26<br><small>0.42</small>       | -0.02<br><small>-0.01</small>     | 0.44<br><small>0.51</small>       | -0.49<br><small>-0.56</small>     | 0.51<br><small>0.57</small>       |
| Total Cloud Cover [frac]                 | 0.64<br><small>0.68</small>       | 0.70<br><small>0.68</small>       | 0.57<br><small>0.62</small>       | 0.74<br><small>0.79</small>       | 0.65<br><small>0.67</small>       | 0.75<br><small>0.69</small>       | 0.57<br><small>0.62</small>       | 0.71<br><small>0.77</small>       | 0.64<br><small>0.67</small>       | 0.66<br><small>0.66</small>       | 0.57<br><small>0.62</small>       | 0.78<br><small>0.79</small>       |
| Low Cloud Cover [frac]                   | 0.34<br><small>0.38</small>       | 0.43<br><small>0.44</small>       | 0.22<br><small>0.27</small>       | 0.48<br><small>0.56</small>       | 0.35<br><small>0.39</small>       | 0.46<br><small>0.49</small>       | 0.21<br><small>0.26</small>       | 0.50<br><small>0.53</small>       | 0.34<br><small>0.38</small>       | 0.44<br><small>0.40</small>       | 0.23<br><small>0.27</small>       | 0.47<br><small>0.57</small>       |
| Medium Cloud Cover [frac]                | 0.31<br><small>0.25</small>       | 0.40<br><small>0.32</small>       | 0.19<br><small>0.16</small>       | 0.48<br><small>0.35</small>       | 0.32<br><small>0.26</small>       | 0.49<br><small>0.38</small>       | 0.19<br><small>0.17</small>       | 0.42<br><small>0.31</small>       | 0.30<br><small>0.24</small>       | 0.30<br><small>0.26</small>       | 0.19<br><small>0.17</small>       | 0.51<br><small>0.38</small>       |
| High Cloud Cover [frac]                  | 0.41<br><small>0.34</small>       | 0.42<br><small>0.34</small>       | 0.39<br><small>0.34</small>       | 0.44<br><small>0.34</small>       | 0.41<br><small>0.34</small>       | 0.46<br><small>0.34</small>       | 0.40<br><small>0.35</small>       | 0.38<br><small>0.31</small>       | 0.41<br><small>0.34</small>       | 0.37<br><small>0.33</small>       | 0.38<br><small>0.33</small>       | 0.51<br><small>0.36</small>       |
| Precipitation (ocean) [Sv]               | 11.80<br><small>13.39</small>     | 1.70<br><small>2.10</small>       | 7.24<br><small>7.86</small>       | 2.86<br><small>3.44</small>       | 11.84<br><small>13.31</small>     | 2.22<br><small>2.56</small>       | 7.33<br><small>7.88</small>       | 2.30<br><small>2.88</small>       | 12.10<br><small>13.48</small>     | 1.21<br><small>1.67</small>       | 7.48<br><small>7.88</small>       | 3.41<br><small>3.93</small>       |
| Precip. minus Evap. (ocean) [Sv]         | -1.89<br><small>-1.28</small>     | 0.25<br><small>0.33</small>       | -2.60<br><small>-2.25</small>     | 0.46<br><small>0.62</small>       | -2.24<br><small>-1.46</small>     | 0.17<br><small>0.18</small>       | -2.75<br><small>-2.24</small>     | 0.33<br><small>0.57</small>       | -1.54<br><small>-1.22</small>     | 0.46<br><small>0.66</small>       | -2.64<br><small>-2.58</small>     | 0.64<br><small>0.68</small>       |
| Precipitation (land) [Sv]                | 4.14<br><small>3.44</small>       | 1.08<br><small>1.07</small>       | 2.91<br><small>2.19</small>       | 0.15<br><small>0.19</small>       | 4.01<br><small>3.25</small>       | 0.74<br><small>0.80</small>       | 3.12<br><small>2.29</small>       | 0.15<br><small>0.16</small>       | 4.29<br><small>3.81</small>       | 1.55<br><small>1.45</small>       | 2.61<br><small>2.16</small>       | 0.14<br><small>0.20</small>       |
| Precip. minus Evap. (land) [Sv]          | 1.85<br><small>1.41</small>       | 0.40<br><small>0.46</small>       | 1.37<br><small>0.84</small>       | 0.07<br><small>0.11</small>       | 2.17<br><small>1.70</small>       | 0.59<br><small>0.73</small>       | 1.54<br><small>0.92</small>       | 0.04<br><small>0.05</small>       | 1.44<br><small>1.18</small>       | 0.17<br><small>0.09</small>       | 1.18<br><small>0.93</small>       | 0.10<br><small>0.17</small>       |
| TOA Net [W/m2]                           | 6.80<br><small>0.90</small>       | -37.73<br><small>-43.38</small>   | 46.82<br><small>45.03</small>     | -32.22<br><small>-43.10</small>   | 14.23<br><small>7.92</small>      | -119.66<br><small>-128.23</small> | 51.71<br><small>50.40</small>     | 69.87<br><small>59.10</small>     | -2.74<br><small>-7.50</small>     | 49.77<br><small>46.85</small>     | 29.23<br><small>27.81</small>     | -121.97<br><small>-132.48</small> |
| TOA SW Net [W/m2]                        | 239.71<br><small>241.34</small>   | 175.55<br><small>180.65</small>   | 302.92<br><small>305.53</small>   | 171.92<br><small>173.64</small>   | 244.65<br><small>245.58</small>   | 72.24<br><small>74.74</small>     | 305.89<br><small>309.38</small>   | 289.22<br><small>288.82</small>   | 233.44<br><small>236.51</small>   | 287.34<br><small>293.85</small>   | 286.29<br><small>289.48</small>   | 69.24<br><small>73.23</small>     |
| TOA LW Net [W/m2]                        | -232.91<br><small>-240.44</small> | -213.27<br><small>-224.03</small> | -256.10<br><small>-260.50</small> | -204.14<br><small>-216.75</small> | -230.43<br><small>-237.66</small> | -191.91<br><small>-202.97</small> | -254.19<br><small>-258.98</small> | -219.35<br><small>-229.73</small> | -236.18<br><small>-244.01</small> | -237.57<br><small>-247.00</small> | -257.06<br><small>-261.67</small> | -191.21<br><small>-205.71</small> |
| TOA SW Net (clear sky) [W/m2]            | 285.30<br><small>287.05</small>   | 218.23<br><small>224.07</small>   | 346.61<br><small>348.53</small>   | 224.38<br><small>227.04</small>   | 293.49<br><small>295.12</small>   | 90.58<br><small>95.21</small>     | 351.83<br><small>353.90</small>   | 374.62<br><small>377.49</small>   | 278.81<br><small>280.40</small>   | 360.06<br><small>363.27</small>   | 329.29<br><small>331.56</small>   | 92.19<br><small>95.19</small>     |
| TOA LW Net (clear sky) [W/m2]            | -258.92<br><small>-268.35</small> | -238.09<br><small>-248.53</small> | -281.87<br><small>-290.60</small> | -231.85<br><small>-243.67</small> | -255.73<br><small>-265.44</small> | -215.80<br><small>-226.83</small> | -280.81<br><small>-290.10</small> | -243.33<br><small>-254.74</small> | -262.70<br><small>-271.79</small> | -262.14<br><small>-271.79</small> | -282.25<br><small>-290.62</small> | -222.44<br><small>-234.14</small> |
| SW Cloud Forcing [W/m2]                  | -45.59<br><small>-45.70</small>   | -42.69<br><small>-43.42</small>   | -43.70<br><small>-43.00</small>   | -52.46<br><small>-53.40</small>   | -48.84<br><small>-49.54</small>   | -18.33<br><small>-20.46</small>   | -45.94<br><small>-44.52</small>   | -85.40<br><small>-88.67</small>   | -45.37<br><small>-43.88</small>   | -72.72<br><small>-69.42</small>   | -43.00<br><small>-42.08</small>   | -22.95<br><small>-21.95</small>   |
| LW Cloud Forcing [W/m2]                  | 26.01<br><small>27.91</small>     | 24.82<br><small>24.50</small>     | 25.77<br><small>30.10</small>     | 27.71<br><small>26.92</small>     | 25.31<br><small>27.78</small>     | 23.89<br><small>23.86</small>     | 26.62<br><small>31.12</small>     | 23.98<br><small>25.01</small>     | 26.52<br><small>27.78</small>     | 24.57<br><small>24.79</small>     | 25.19<br><small>28.95</small>     | 31.23<br><small>28.43</small>     |
| Surface Net SW [W/m2]                    | 161.67<br><small>160.00</small>   | 110.91                            | 209.28                            | 113.04                            | 166.61                            | 44.84                             | 212.49                            | 192.59                            | 154.75                            | 181.17                            | 195.17                            | 43.96                             |
| Surface Net LW [W/m2]                    | -58.57<br><small>-56.00</small>   | -54.99                            | -63.88                            | -51.08                            | -57.88                            | -52.51                            | -64.16                            | -50.14                            | -58.18                            | -53.89                            | -63.16                            | -52.09                            |
| Surface SH [W/m2]                        | -17.52<br><small>-21.00</small>   | -19.65                            | -20.31                            | -9.56                             | -16.46                            | -18.80                            | -19.92                            | -6.88                             | -19.17                            | -22.36                            | -20.68                            | -12.82                            |
| Surface LH [W/m2]                        | -78.45<br><small>-82.00</small>   | -42.75                            | -109.35                           | -49.65                            | -78.12                            | -43.89                            | -112.03                           | -41.55                            | -80.92                            | -42.62                            | -111.00                           | -56.42                            |
| Net Surface (no snowfall) [W/m2]         | 7.13<br><small>0.60</small>       | -6.47                             | 15.74                             | 2.75                              | 14.16                             | -70.36                            | 16.39                             | 94.03                             | -3.52                             | 62.30                             | 0.33                              | -77.37                            |
| Net Surface [W/m2]                       | 6.37<br><small>0.60</small>       | -7.85                             | 15.73                             | 1.07                              | 13.31                             | -72.91                            | 16.37                             | 93.13                             | -4.14                             | 62.06                             | 0.33                              | -79.64                            |
| TOA - Sfc Net Radiation (no snow) [W/m2] | -0.33<br><small>0.00</small>      | -31.26                            | 31.08                             | -34.97                            | 0.06                              | -49.30                            | 35.31                             | -24.16                            | 0.78                              | -12.53                            | 28.90                             | -44.60                            |
| TOA - Sfc Net Radiation [W/m2]           | 0.43<br><small>0.00</small>       | -29.87                            | 31.09                             | -33.29                            | 0.92                              | -46.75                            | 35.34                             | -23.26                            | 1.40                              | -12.29                            | 28.90                             | -42.33                            |
|                                          | ALL Global                        | ALL North Midlat                  | ALL Tropical                      | ALL South Midlat                  | DJF Global                        | DJF North Midlat                  | DJF Tropical                      | DJF South Midlat                  | JJA Global                        | JJA North Midlat                  | JJA Tropical                      | JJA South Midlat                  |

Model Bias  
(standard deviation of interannual variability from observations)

10  
9  
8  
7  
6  
5  
4  
3  
2  
1  
0  
-1  
-2  
-3  
-4  
-5  
-6  
-7  
-8  
-9  
-10